Assignment 1: How Matrix Size Affects the Runtime and Memory of Basic Matrix Multiplication in Python, Java and C

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Code, data and instructions: `https://github.com/Perdomin38/matmul-basic-benchmark`

October 2026

Abstract

This study asks how the matrix size n affects the runtime and the memory use of the same textbook $O(n^3)$ matrix multiplication written in Python, Java and C. The three implementations use the same triple loop (i–j–k order), the same float64 inputs read from shared files, and they were validated against known answers and against NumPy before benchmarking. On one laptop (Intel Core i7-13620H, Windows 11), every configuration was run five times after warm-up, and the median, the interquartile range and the peak process memory were recorded for $n = 100, 200, \ldots, 3200$. Runtime grew by roughly 8 to 12 times per doubling of n while the data fitted in the CPU caches, as the cubic model predicts, and grew faster for larger matrices. C was the fastest implementation at every tested size; Java took 1.2 to 1.6 times as long as C up to $n = 800$ but 6.5 times as long at $n = 1600$; CPython took 87 to 162 times as long as C. The largest sizes that finished within a 600 s budget were $n = 1600$ for C and Java and $n = 800$ for Python. These conclusions apply only to these implementations, settings and this machine.

1 Introduction and objective

Matrix multiplication is a basic building block of data science and machine learning, and its cost grows quickly with the size of the matrices. Before studying optimised, sparse, parallel and distributed versions in later assignments, this first study establishes a *baseline*: the simplest algorithm written in three widely used languages that execute code in very different ways (an interpreter, a JIT-compiling virtual machine and an ahead-of-time compiler).

Research question. *How does the matrix size n affect the runtime and the memory use of the basic triple-loop matrix multiplication implemented in Python, Java and C on one fixed machine?*

The factor that changes is n (side length of the $n \times n$ square matrices) and the implementation language; the outcomes are the multiplication time and the peak memory of the process.

Hypotheses (predictions made before measuring).

H1 Runtime grows approximately as n^3, i.e. about 8 times longer every time n doubles, for all three implementations.

H2 C and Java will be within a small factor of each other, and CPython will be one to two orders of magnitude slower, because it interprets every operation.

H3 Memory grows approximately as n^2 (three $n \times n$ matrices), with a constant runtime overhead that is largest for the JVM and with a larger per-element cost in Python, where every number is an object.

2 Background and references

Algorithm. For $C = AB$ with $A, B \in \mathbb{R}^{n \times n}$, each element is the dot product $c_{ij} = \sum_{k=0}^{n-1} a_{ik} b_{kj}$. Computing n^2 elements, each with n multiply–add operations, costs exactly n^3 multiplications and n^3 additions, so the work is $\Theta(n^3)$ [1]. Asymptotically faster algorithms exist (e.g. Strassen) but are out of scope; the assignment requires the straightforward triple loop.

Memory hierarchy. The time of a multiply–add depends on where its operands are: registers and caches are much faster than main memory [2]. In the i–j–k order, the inner loop reads a *row* of A (consecutive addresses, cache friendly) but a *column* of B (addresses $8n$ bytes apart, one cache line per element). When the matrices no longer fit in the caches, the cost per operation is expected to rise, so measured times can grow faster than the pure n^3 model.

Execution models. C is compiled ahead of time to machine code by GCC. Java is compiled to bytecode, which the HotSpot JVM first interprets and then compiles to machine code just-in-time (JIT) once a method is "hot"; this is why warm-up runs are needed [4]. CPython interprets bytecode and represents every float as a heap object, so every arithmetic operation involves type dispatch and object handling.

Benchmarking practice. Repeated measurements, warm-up, and robust summaries (median, interquartile range) are recommended because single measurements of modern systems are noisy [4, 5].

3 Methodology

3.1 Implementations

The three programs implement exactly the same loop (Listing 1), with a local accumulator and the same i–j–k order, using 64-bit IEEE-754 floating point (`double` in C and Java, `float` in Python). No optimised library is used inside any measured kernel.

Listing 1: Kernel shared by the three implementations (C version).

```
for (int i = 0; i < n; i++)
    for (int j = 0; j < n; j++) {
        double sum = 0.0;
        for (int k = 0; k < n; k++)
            sum += A[i * n + k] * B[k * n + j];
        C[i * n + j] = sum;
    }
```

Storage layouts (unavoidable language differences).

- **C**: one contiguous `malloc` block of n^2 doubles per matrix, row-major (`A[i*n+k]`); 8 bytes per element.
- **Java**: `double[n][n]`, an array of n row arrays. Each row is contiguous, but rows are separate heap objects, and each access `B[k][j]` first loads the reference to row `k` and is bounds-checked.
- **Python**: a list of n lists. Each list stores 8-byte pointers to separate `float` objects (24 bytes each in CPython), so about 32 bytes per element, possibly scattered in memory.

3.2 Inputs

For every n, the matrices A and B were generated once with NumPy (`default_rng(2026 + n)`, uniform values in $[-1, 1)$) and saved as binary files of little-endian float64 values in row-major order. All three programs read the same files, so they multiply exactly the same logical matrices and no cross-language random generator is involved. Declared size sequence: $n = 100, 200, 400, 800, 1600, 3200$; doubling n makes the expected cubic growth easy to check (8 times longer per step).

3.3 Environment and fixed settings

Item	Value
CPU	Intel Core i7-13620H: 6 performance + 4 efficiency cores, 16 logical CPUs 24 MB shared L3 cache [6]
RAM / OS	16 GB / Windows 11 Home (build 26200), laptop plugged in
C	GCC 16.2.0 (MinGW-w64 UCRT), flags `-O2 -std=c11 -static`
Java	Oracle JDK 24.0.1, HotSpot 64-bit Server VM, flag `-Xmx3g`
Python	CPython 3.11.9 (standard library only in the kernel)
Helper tools	NumPy 2.2.0 (inputs, reference), psutil 6.1.0 (memory), pandas, matplotlib
Threads	all kernels single-threaded; no numerical library inside the kernels

Table 1: Machine and toolchain. These settings were fixed for all runs.

The flag `-O2` is GCC's standard optimisation level; it was kept fixed and no architecture-specific (`-march=native`) or unsafe floating-point (`-ffast-math`) flags were used, so the compiler may not reorder the floating-point sum. `-static` only avoids a run-time DLL dependency. The JVM used its default JIT (tiered compilation) and garbage collector (G1); `-Xmx3g` caps the heap below the memory budget.

Because the CPU is hybrid, a first exploratory run (kept in `results/exploratory_unpinned/`) showed large run-to-run variation, e.g. five consecutive Python runs at $n = 200$ ranged from 0.67 s to 2.95 s, consistent with the operating system moving the process between fast and slow cores. In the final run every benchmark process was therefore pinned to the performance cores (logical CPUs 0–11) with high priority.

3.4 Timing protocol

- **Timed region**: only the multiplication kernel. Reading the input files and allocating the result matrix happen *before* the timer starts.
- **Timers**: monotonic clocks: `clock_gettime(CLOCK_MONOTONIC)` in C, `System.nanoTime()` in Java, `time.perf_counter()` in Python.
- **Warm-up**: C 1 run, Python 1 run (caches, page faults), Java 2 runs (JIT compilation). Warm-up runs are recorded in the raw data but excluded from the statistics.
- **Repetitions**: 5 measured runs per configuration, all in the same process. Reported: median and interquartile range (IQR $= Q_3 - Q_1$), plus min and max.
- **Result consumption**: after each run the program sums all elements of C (checksum) and prints it, so the computation cannot be skipped by the compiler; each checksum is compared with NumPy's result (relative tolerance 10^{-9}).

3.5 Memory measurement

The metric is the **peak resident memory of the whole process** (on Windows, the *peak working set*), read with psutil [7] every 10 ms by the benchmark script. It includes the language runtime (JVM, interpreter) and all data, not only the matrices. As a reference, the raw storage of three float64 matrices is $3 \times 8n^2$ bytes.

3.6 Budgets and limits

Before the large tests, a budget was declared per configuration (one language and one n, including warm-ups): **600 s of wall time** and **4096 MB of memory**. A configuration exceeding a budget is killed and recorded as TIMEOUT or MEMORY_LIMIT; larger sizes of that language are then recorded as SKIPPED. No execution time is invented for them.

3.7 Correctness checks

Before benchmarking, each implementation multiplied four test cases and wrote its result to a file: (1) a hand-computed 2×2 case $\left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)\left(\begin{smallmatrix}5&6\\7&8\end{smallmatrix}\right) = \left(\begin{smallmatrix}19&22\\43&50\end{smallmatrix}\right)$; (2) $A \cdot I = A$ with $n = 37$; (3) $A \cdot 0 = 0$ with $n = 37$ (all three required *exact* equality); and (4) random A, B with $n = 100$ compared with NumPy's `A @ B`, accepted if $|c - c_{\text{ref}}| \leq 10^{-9} + 10^{-9}|c_{\text{ref}}|$ for every element. The tolerance is justified by the rounding-error bound of a dot product, $|\hat{c} - c| \leq \gamma_n \sum_k |a_{ik} b_{kj}|$ with $\gamma_n \approx nu$ and $u \approx 1.1 \times 10^{-16}$ [3]: for $n = 100$ and values in $[-1, 1)$ this is of order 10^{-13}, far below 10^{-9}, while an indexing bug would produce errors of order 1. NumPy sums in a different order (BLAS), so tiny differences are legitimate.

4 Results

4.1 Correctness

All implementations passed all tests (Table 2). The exact tests gave zero error. In the random test the maximum difference from NumPy was 5.3×10^{-15}, and the three implementations produced *bitwise identical* matrices, as expected since they perform the same operations in the same order with IEEE-754 doubles. During the benchmark, every checksum matched the NumPy reference.

Test	n	Criterion	Max abs. error (C, Java, Python)	Passed
known 2×2	2	exact	0, 0, 0	yes (all)
identity $A \cdot I$	37	exact	0, 0, 0	yes (all)
zero $A \cdot 0$	37	exact	0, 0, 0	yes (all)
random vs NumPy	100	rtol = atol = 10^{-9}	5.3×10^{-15} (each)	yes (all)

Table 2: Correctness tests (`results/raw/correctness.csv`).

4.2 Runtime

Table 3 summarises the five measured runs of every configuration that finished within budget; Figure 1 shows the medians on log–log axes.

Lang.	n	Median (s)	IQR (s)	IQR/median	ns per mult-add	Growth (times)
C	100	0.0002703	0.000011	4.0%	0.27	—
C	200	0.002717	0.000091	3.3%	0.34	10.1
C	400	0.03216	0.00018	0.6%	0.50	11.8
C	800	0.3357	0.0026	0.8%	0.66	10.4
C	1600	4.122	0.23	5.5%	1.01	12.3
Java	100	0.0004409	0.0000037	0.8%	0.44	—
Java	200	0.003361	0.000028	0.8%	0.42	7.6
Java	400	0.05294	0.0088	16.6%	0.83	15.8
Java	800	0.4976	0.058	11.6%	0.97	9.4
Java	1600	26.79	1.9	7.0%	6.54	53.8
Python	100	0.03706	0.0014	3.7%	37.06	—
Python	200	0.3091	0.0034	1.1%	38.64	8.3
Python	400	2.803	0.029	1.0%	43.80	9.1
Python	800	54.30	2.2	4.1%	106.06	19.4

Table 3: Multiplication time, 5 measured runs per row (warm-ups excluded). "ns per mult-add" = median/n^3; a constant value means perfect cubic scaling. "Growth" = median at n divided by median at $n/2$ (cubic model: 8). Source: `results/tables/runtime_summary.csv`.

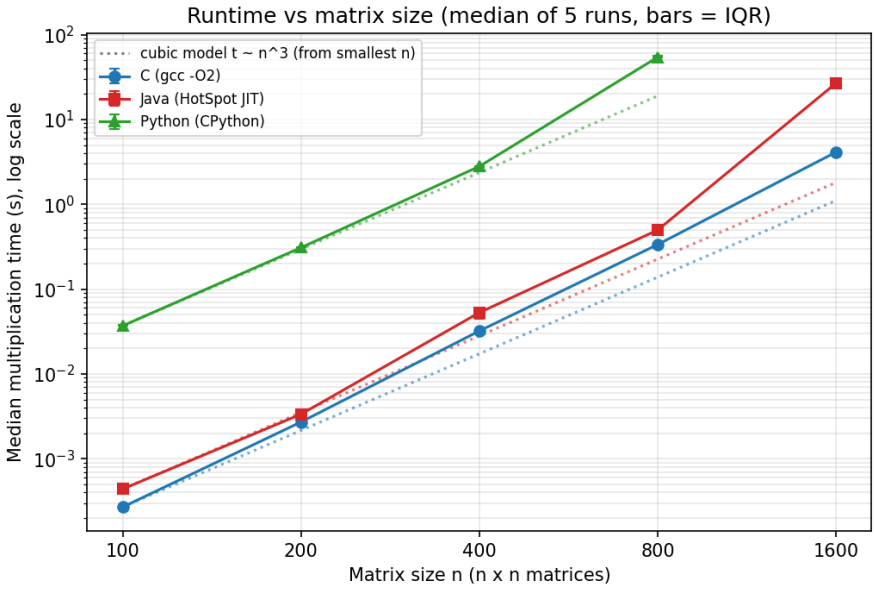


Figure 1: Median multiplication time versus n (log–log). Error bars show the IQR (mostly smaller than the markers). Dotted lines extrapolate each language's smallest-n time with the cubic model $t \propto n^3$.

Ratios of medians with respect to C: Java/C = 1.6, 1.2, 1.6, 1.5, 6.5 and Python/C = 137, 114, 87, 162 for $n = 100, 200, 400, 800, (1600)$ respectively. Fitting $t = an^b$ to all completed sizes gives $b = 3.47$ (C), 3.90 (Java) and 3.47 (Python). Figure 2 shows the cost per inner-loop operation.

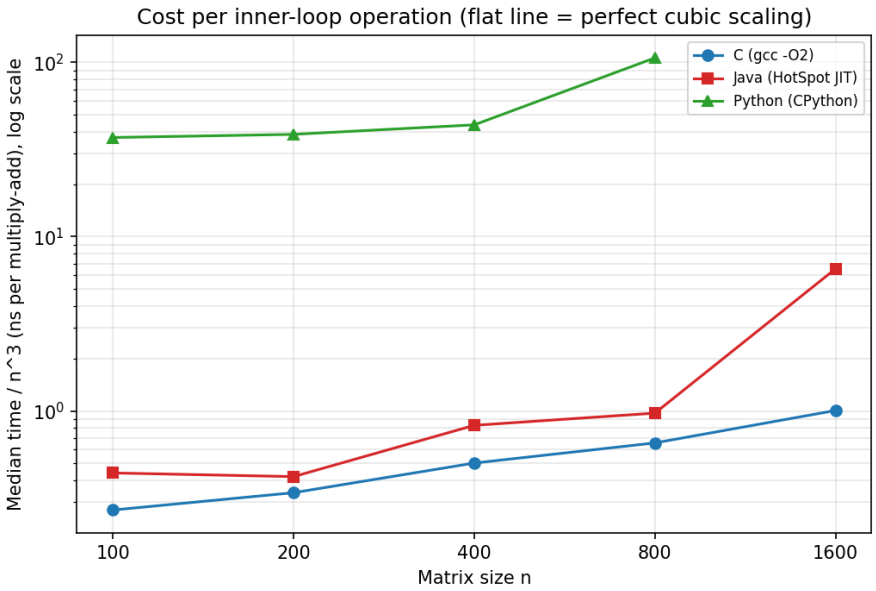


Figure 2: Median time divided by n^3 (nanoseconds per multiply–add). A flat line would mean exact cubic scaling; an increasing line means each operation becomes more expensive as the matrices grow.

Warm-up. For Java, the first warm-up run took 5.7 times ($n = 100$) and 2.4 times ($n = 200$) as long as the measured median, consistent with the JIT compiling the kernel during the first runs. From $n = 400$ the ratio was close to 1, since the JIT compiles the loop early inside the (longer) first run. For C and Python the first run was within 6% of the median (27% for C at $n = 100$, where a run lasts 0.3 ms).

4.3 Memory

n	C (MB)	Java (MB)	Python (MB)	$3 \times 8n^2$ bytes (MB)
100	9.1	43.0	16.5	0.23
200	9.0	46.5	20.3	0.92
400	11.8	52.5	33.8	3.66
800	22.8	75.0	89.6	14.65
1600	66.7	198.4	—	58.59

Table 4: Peak resident memory (peak working set) of the whole process, measured with psutil; MB $= 2^{20}$ bytes. Source: `results/tables/memory_summary.csv`.

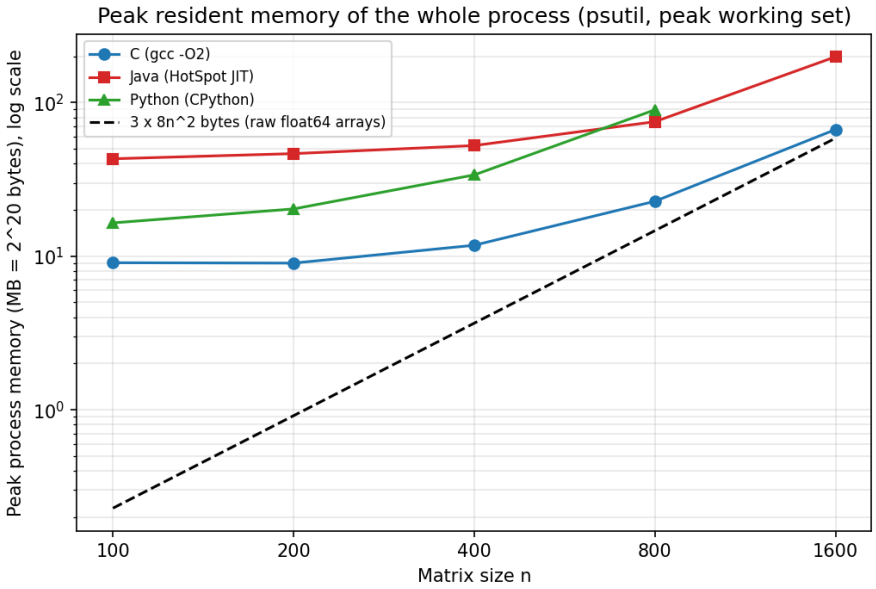


Figure 3: Peak process memory versus n (log–log). The dashed line is the raw storage of three float64 matrices.

4.4 Limits

Language	Largest feasible n	First size over budget	Outcome at larger sizes
C	1600	3200 (TIMEOUT after 600 s, peak 242.5 MB)	—
Java	1600	3200 (TIMEOUT after 600 s, peak 491.2 MB)	—
Python	800	1600 (TIMEOUT after 600 s, peak 311.7 MB)	3200 SKIPPED

Table 5: Largest size that met the 600 s / 4096 MB budget. No configuration reached the memory budget. Source: `results/raw/runs_status.csv`.

The timed-out configurations are not summarised, but the runs that completed before the limit are kept in the raw data: at $n = 3200$ a single C multiplication took about 252 s, at least 60 times the $n = 1600$ median (cubic model: 8 times); a Java warm-up run took 309 s; and at $n = 1600$ one Python warm-up run took 336 s.

5 Discussion

Answer to the question (H1). While the data are small, the runtime follows the cubic model closely: Python's time grows 8.3 and 9.1 times per doubling up to $n = 400$, and C's and Java's grow 7.6 to 12 times up to $n = 800$. The cost per operation is not perfectly constant, however (Figure 2): for C it increases smoothly from 0.27 ns to 1.01 ns, so the fitted exponents ($b \approx 3.5$–3.9) are larger than 3. The cubic model counts operations correctly, but it ignores how expensive each operation is, which depends on the memory system and on the machine.

Memory-layout effects. The sharpest changes appear when the working data outgrow the 24 MB L3 cache. Three matrices of $n = 800$ take 14.6 MB in C and Java, but about $3 \times 32 \times 800^2 \approx 61$ MB in Python; Python's cost per operation jumps from 43.8 ns to 106 ns exactly at $n = 800$. At $n = 1600$ (58.6 MB of raw data) the cost per operation of Java jumps from 0.97 ns to 6.5 ns, whereas C, running the same loop on the same inputs, increases only from 0.66 ns to 1.01 ns. Because the algorithm, inputs and machine are the same, the most plausible cause is the storage layout: walking down a column of B in C uses addresses with a constant stride inside one block, which the hardware can predict, while in Java each step must first load a different row reference and the rows are separate objects. C shows a similar cliff only one size later (about 252 s at $n = 3200$, at least 60 times the time at $n = 1600$). These explanations are consistent with the evidence but were not confirmed with hardware counters (cache or TLB misses), so they remain hypotheses.

Interpreter and runtime effects (H2). H2 is supported. Up to $n = 800$ Java takes only 1.2 to 1.6 times as long as C: after JIT compilation both execute machine code, and the remaining gap is consistent with bounds checks and the extra row indirection. CPython takes 87 to 162 times as long as C, because every `+` and `*` is executed by the interpreter on boxed `float` objects, which costs about 37–44 ns per multiply–add instead of below 1 ns. The JIT is also visible in Java's warm-up runs.

Memory (H3). H3 is supported. The runtime overhead dominates for small n: the JVM needs about 43 MB before the matrices matter, Python about 16 MB, C about 9 MB. For large n the growth follows n^2. C's increase from $n = 100$ to 1600 (57.6 MB) matches the raw storage (58.4 MB). Python's increase from $n = 100$ to 800 (73 MB for 1.92 million elements) is about 40 bytes per element, i.e. about 5 times the 8 bytes of a C double, as expected from pointer plus float object. Java at $n = 1600$ (198 MB) uses more than three times the raw storage; the JVM heap also holds the byte buffers used while reading the input files until the garbage collector reclaims them, and the heap grows in steps. Memory never limited the experiment: time did.

System effects and limitations.

- **One machine.** All results come from one laptop; with a hybrid CPU, unpinned runs were unreliable (Section 3.3), so pinning to performance cores was part of the protocol.
- **Sustained load.** A supplementary check (`results/raw/supplementary_sustained_load_check.txt`) ran the Python $n = 800$ case four times in a row: 31.9, 40.9, 43.4 and 57.2 s, independently of pinning. The time increases with sustained load, which is consistent with the laptop lowering its clock under sustained load (this CPU has a lower sustained than short-term power limit [6]). The frequency was not logged, so this is an inference. Long runs (large n) may therefore run at a lower clock than short runs, which can contribute to the increase of the cost per operation. Even the fastest of these runs (31.9 s) means about 62 ns per multiply–add for Python at $n = 800$, still above the 44 ns measured at $n = 400$. The jump of Java at $n = 1600$ is still significant because C at the same size, under the same conditions, did not show it.
- **Few repetitions for long runs.** Five repetitions give a rough IQR; Java at $n = 400$ and 800 showed 12–17% IQR (the first measured runs were faster than the later ones).
- **Memory metric.** Peak working set is sampled every 10 ms and includes the whole runtime; it is not comparable to an array-size estimate or to a managed-heap measurement.
- **Coarse size grid.** Doubling n cannot locate exactly where the cliffs start.

6 Conclusions

Within the tested scope (these three basic implementations, i–j–k order, float64, GCC `-O2`, HotSpot JDK 24, CPython 3.11, one i7-13620H laptop with pinning to performance cores):

- The runtime of all three implementations follows the cubic model (about 8 times longer per doubling) while the data fit in cache, and grows faster beyond that; the operation count alone does not predict the time.
- C was the fastest at every size. Java took at most 1.6 times as long as C up to $n = 800$ but 6.5 times as long at $n = 1600$; CPython was about two orders of magnitude slower than C.
- Memory grows as n^2 plus a runtime-dependent constant; Python needs about 5 times more memory per element than C.
- The largest feasible sizes under the 600 s budget were $n = 1600$ (C, Java) and $n = 800$ (Python).

This study does not show that one language is universally faster: the gap depends on the size, on the storage layout and on the machine. What remains unresolved is the exact cause of the large-n cliffs (cache versus TLB versus clock frequency), which would require hardware counters and frequency logging; optimised loop orders and storage layouts are left for Assignment 2.

Reproducibility materials

Code, raw data (CSV), analysis scripts and exact instructions are in the repository `https://github.com/Perdomin38/matmul-basic-benchmark` (see `README.md`). All fixed settings are in `scripts/common.py`. The pipeline is: `build.py` → `generate_inputs.py` → `verify.py` → `run_benchmarks.py` → `analyze.py`. Raw measurements: `results/raw/benchmark_raw.csv` (every warm-up and measured run) and `results/raw/runs_status.csv`.

Use of AI tools

An AI assistant (Claude, Anthropic) was used to help plan the experimental design, write and debug the scripts, and revise the English of this report. All measurements were obtained by running the code on the machine described above, and the author has checked the code, the results and the references and is responsible for the conclusions.

References

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